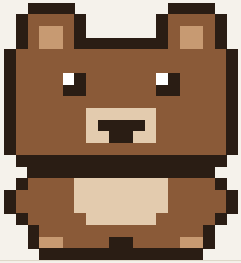
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 7: Instrumental variables, regression discontinuity and synthetic control



Dr. Matteo Caruso · mcaruso4@os.uniroma3.it
Roma Tre University – Department of Business Economics
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Where we are

- Monday: difference-in-differences, a comparison group followed over time
- Today, three designs, at speed:
  - **Instrumental variables**: something outside the firm's control pushes some units into treatment
  - **Regression discontinuity**: a cut-off decides who is treated
  - **Synthetic control**: only one unit is treated, so we build its twin

Design	The comparison group	Key assumption
Instrumental variables	units the outside factor did not push	the factor works only through the treatment
Regression discontinuity	units just below the cut-off	no sorting around the cut-off
Synthetic control	a weighted mix of untreated units	the mix tracked the treated unit for years before

PART 1

Instrumental variables



You have already used an instrument

Lecture 5, our grant lottery:

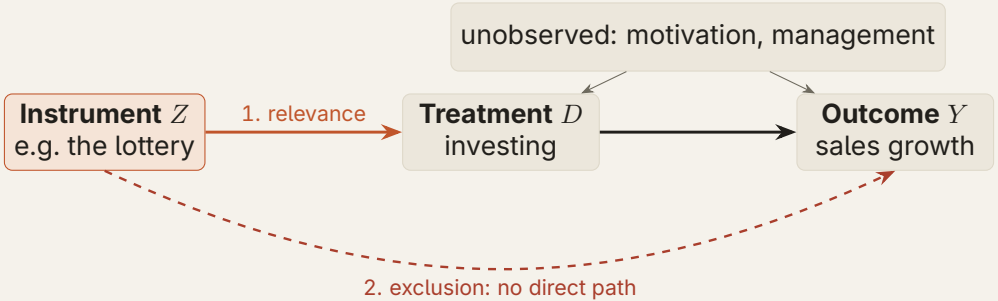
- Winning the lottery is **random**, but only 60% of winners actually invest
- Comparing winners and losers gives the effect of the **offer**: ITT $= 1.8$ points
- The effect on firms that **invest** because they won:

$$\frac{\text{effect of winning on sales growth}}{\text{effect of winning on investing}} = \frac{1.8}{0.60} = 3 \text{ points}$$

> The lottery is an **instrument**: it changes who gets treated, at random, and has no other way to affect the outcome

Instrumental variables generalise this ratio to situations where nobody ran a lottery, but something else played its role.

The idea



1. **Relevance**: Z clearly changes the treatment (checkable in the data)
2. **Exclusion**: Z affects the outcome **only** through the treatment (not checkable: an argument)
3. Z is as good as random: unrelated to motivation, ability, management

The Wald ratio

EFFECT OF THE TREATMENT

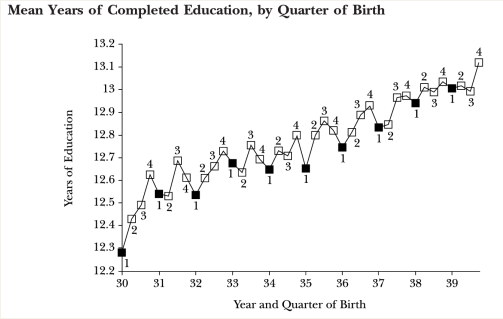
$$\rho = \frac{\text{difference in outcomes between } Z = 1 \text{ and } Z = 0}{\text{difference in treatment between } Z = 1 \text{ and } Z = 0} = \frac{\text{reduced form}}{\text{first stage}}$$

- **First stage**: how much the instrument moves the treatment. If it barely moves it (a **weak** instrument), we divide by almost zero and the estimate is unreliable
- **Reduced form**: how much the instrument moves the outcome
- With controls and several instruments, software runs “two-stage least squares”: same logic, same ratio

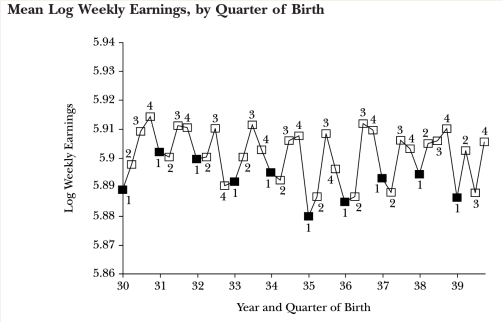
A classic: quarter of birth and schooling (Angrist and Krueger, 1991)

- Question: what is the return to one more year of school? Comparing educated and less educated people mixes school with ability and family background
- In the US, children start school in the calendar year they turn 6, and may leave at 16
- Those born early in the year reach 16 after fewer years of school: if they leave as soon as allowed, they have **slightly less schooling**
- The month of birth is unrelated to ability or family: a natural instrument

Quarter of birth: first stage and reduced form



First stage: schooling by quarter of birth.



Reduced form: earnings by quarter of birth.

Men born early in the year have slightly less schooling **and** slightly lower earnings: if the only link is schooling, the ratio of the two gaps is the return to school

The result

	Born Q1	Born Q4	Difference
log weekly wage	5.892	5.905	-0.0135
years of school	12.688	12.839	-0.151
Wald estimate: $-0.0135 / -0.151$			**0.089**
OLS (simple comparison)			0.070

Men born 1930–39, 1980 US census. Angrist and Krueger (1991), as reported in Angrist and Imbens (1995).

- One more year of school raises earnings by about **9%**
- The whole estimate rests on a tiny difference: 0.15 years of school
- Exclusion: could the month of birth affect earnings in other ways? (age at school entry, season of birth and health)

Whose effect? Compliers

An instrument moves only some units. Four types:

- **Always-takers**: treated anyway (e.g. would invest even without the grant)
- **Never-takers**: never treated (would not invest even with it)
- **Compliers**: treated **because of** the instrument
- Defiers: do the opposite (assumed away)

> **LOCAL AVERAGE TREATMENT EFFECT (LATE)**
>
> IV measures the effect for the **compliers** only. Angrist and Krueger: men who stayed in school because the law kept them there. Our lottery: firms that invested because they won

For policy this is often the relevant group: the firms that a policy actually moves.

Instruments in business and policy

- **Encouragement**: a programme open to all; some firms get a randomly assigned call or visit, the instrument for taking part
- **Rules**: assignment by date or by a randomly assigned official
- **Distance** to a bank or university, for credit or education: exclusion often doubtful

WHERE IV GOES WRONG

- **Weak first stage**: the instrument hardly moves the treatment
- **Exclusion fails**: the instrument has its own path to the outcome. Not testable: it must be argued

PART 2

Regression discontinuity

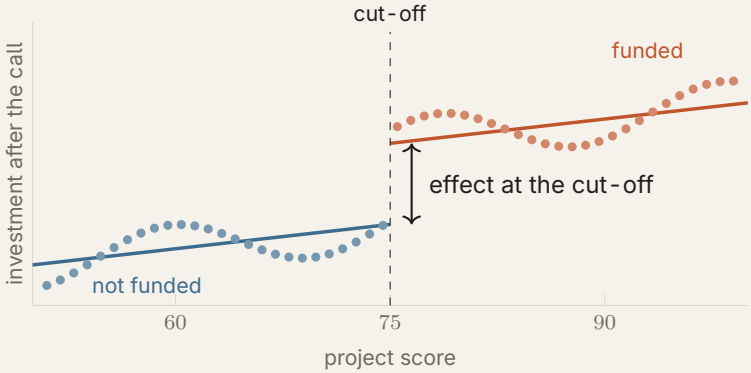


Rules with a cut-off

- Many policies assign treatment by a **score** and a **threshold**:
  - a grant to projects scoring 75 or more
  - a tax rule for firms below 50 employees
  - a subsidy for municipalities under 5,000 inhabitants
- Firms far above and far below the threshold differ a lot
- Firms at 74 and at 76 are almost identical, except that one is funded

> **THE IDEA**
>
> Compare units **just above** and **just below** the cut-off: near the threshold, treatment is as good as random

The picture



Illustrative data. Without the grant, the two lines would join smoothly at 75; the jump is the effect.

What must hold, and how to check it

THE ASSUMPTION

Nothing else jumps at the cut-off, and units cannot **precisely** place themselves on one side

- **Sorting**: firms able to push their score just above the line differ from those just below. Check: suspiciously many units just above?
- **Balance**: size, sector, age (fixed before the call) should not jump at the cut-off
- **Other rules at the same threshold**: if 50 employees triggers other obligations too, the jump mixes them

The price: the effect is **local**. A grant that works for projects scoring 75 may not work for those scoring 95.

R&D grants in Emilia-Romagna (Bronzini and Iachini, 2014)

- 2003: the Region funds industrial research and pre-competitive development; an independent committee scores each project:

technical and scientific	up to 45 points
financial and economic	up to 20 points
managerial	up to 20 points
regional impact	up to 15 points
funded if total $\geq$ 75 (out of 100)	

- A **sharp** RDD: compare investment of firms just above and just below 75
- Firms could hardly control their score. But there are **fewer** projects at 74 than expected: the authors' reading is that the committee avoided scoring a project just below the line

What they found

- All firms together: **no significant increase** in investment. Firms partly replaced their own money with public money (crowding out)
- By size:
  - **small** firms increased investment by about the amount of the grant
  - **larger** firms: no additional investment

> **THE POLICY LESSON**
>
> The same euro has different effects on different firms. A result like this argues for **targeting** the grants on small firms, which are more likely to be short of finance

Remember Lecture 5: a “no effect” on the whole sample can hide a real effect on a subgroup, and vice versa.

A cut-off in time: an unexpected shortage of funds

- de Blasio, Fantino and Pellegrini (2011): a national fund for technological innovation
- At some point the money ran out, unexpectedly. Firms could still apply, hoping funds would return
- Applications were suspended after ten months; funding took five years to come back
- Comparison: firms that applied **just before** the shortage (funded) and **just after** (not funded)

RESULT

The programme did not raise innovative investment: public funding replaced private funding

The cut-off is the date of application, not a score. The key question: did the firms that applied after the shortage differ from those that applied before?

Law 488 again: many cut-offs

- Law 488 ranked applicants by region and call, and funded them until the money ran out: **each ranking has its own cut-off**
- Cerqua and Pellegrini (2014) compare firms just above and just below each cut-off
- They find positive effects on investment, employment and turnover, **not** due to investment brought forward

TWO DESIGNS, TWO ANSWERS

Monday, Bronzini and de Blasio (DiD, all funded vs all rejected): part of the investment was only brought forward. Today, Cerqua and Pellegrini (RDD, near the cut-offs): real additional investment. Different samples, periods and methods: which one would you show a minister, and why?

Sharp and fuzzy: where the two designs meet

- **Sharp** RDD: crossing the cut-off decides treatment for everyone (Emilia-Romagna)
- **Fuzzy** RDD: crossing the cut-off only raises the **probability** of treatment (some funded firms renounce, some below get money elsewhere)

> In a fuzzy RDD, being above the cut-off is an **instrument**: effect = jump in the outcome / jump in the probability of treatment. The same ratio as the lottery and as quarter of birth

PART 3

Synthetic control



When only one unit is treated

- Many policies hit **one** unit: a region passes a law, a country leaves a union, a firm opens in one market
- DiD needs a control group; with one treated region, which region is the right comparison?
- Picking one by hand (“Lombardy is like Piedmont”) is arbitrary

THE IDEA

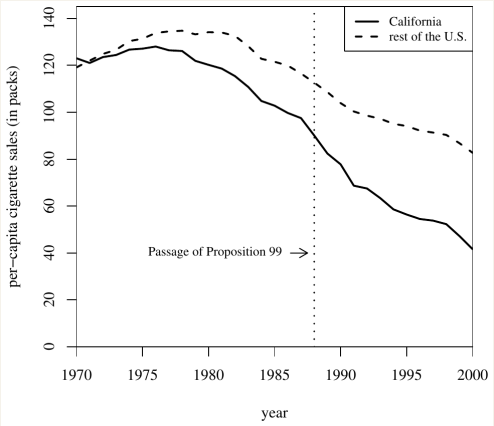
Build a **synthetic twin**: a weighted average of untreated units, with weights chosen so that the twin reproduces the treated unit’s outcome for **many years before** the policy. After the policy, the twin shows what would have happened

The weights are positive and add up to one: the twin is a mix of real units, which can be named and discussed.

California, 1988: Proposition 99

- November 1988: Californian voters approve a tobacco control programme: **+25 cents** of tax per pack, anti-smoking campaigns, local smoke-free rules
- Did it reduce smoking? Per-capita cigarette sales, 1970–2000
- The rest of the US is no counterfactual: sales were falling faster in California already before 1988

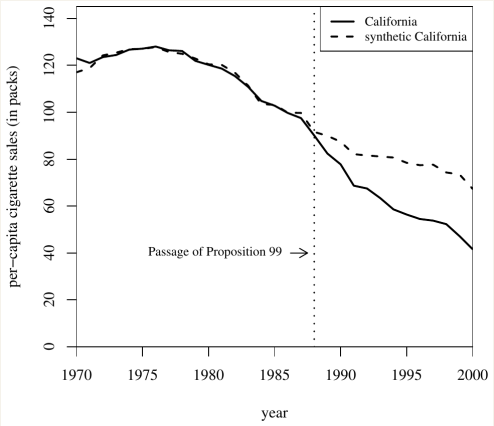
Abadie, Diamond and Hainmueller (2010).



California (solid) vs the rest of the US (dashed).

Synthetic California

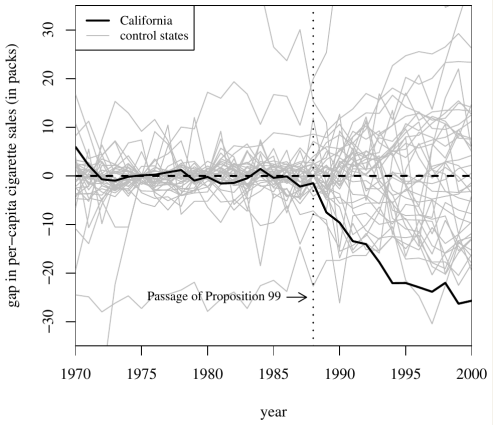
- Donor pool: 38 states, **excluding** those with their own tobacco programmes or large tax rises in the period
- The best mix: **Colorado, Connecticut, Montana, Nevada and Utah**; all others get zero weight
- The twin tracks California almost perfectly for 18 years before 1988
- By 2000, sales in California are about **26 packs per person lower** than in the synthetic California



California (solid) vs synthetic California (dashed).

Is the gap real? Placebos

- With one treated unit there is no usual standard error
- Instead: pretend each donor state had the policy in 1988, build its own synthetic twin, and compute its "effect"
- If California's gap is one among many similar ones, it may be noise
- Here California's gap (black) is among the most extreme: unlikely to be chance



Gap between each state and its synthetic twin; California in black, the 38 placebo states in grey.

An Italian case: the cost of organised crime

- Pinotti (2015): **Apulia and Basilicata**, two southern regions where mafia activity spread after the 1970s
- Synthetic twin: a weighted mix of regions less affected by organised crime, matched on economic structure and outcomes in the years before
- Result: GDP per capita about **16% lower** than in the synthetic twin, while murders rose sharply
- Mechanism suggested by the author: private investment replaced by **less productive public investment**, not a shift to the black economy

> **FOR FIRMS**
>
> The same logic can evaluate a single big decision: a product launched in one region, a plant opened in one country. The twin is a mix of regions or markets where nothing happened

What must hold

- A **long** pre-policy period with a close fit: if the twin cannot track the treated unit before, it cannot be trusted after
- Donors must **not be affected** by the policy (spillovers) or by their own big policies (hence the excluded states)
- No other shock hits the treated unit at the same time

> Strength: transparent and visual, with no econometrics on screen. Weakness: one unit, so every result is a case study

PART 4

Reading these studies



Questions to ask

Instrumental variables

1. Does the instrument clearly move the treatment?
2. Can you think of any other path from the instrument to the outcome?
3. Who are the compliers, and do we care about them?

Regression discontinuity

1. Who sets the score, and could units manipulate it?
2. Does anything else change at the cut-off?
3. Is the effect near the cut-off the one the policy question needs?

Synthetic control

1. How well does the twin fit before the policy, and for how long?
2. Which units make up the twin, and are they plausible?
3. Were the donors touched by the policy or by their own shocks?

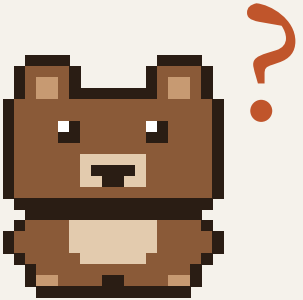
All three designs buy credibility with **locality**: a convincing effect for a specific group, not for everyone

Next

- **Friday**: lab and discussion
- **Next week**: ex post evaluation as an approach: evaluation questions, theory of change, and reading an evaluation critically. End of Part I
- Readings on the Lectures page, week 4

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- de Blasio, G., D. Fantino and G. Pellegrini (2011), Evaluating the impact of innovation incentives: evidence from an unexpected shortage of funds, Banca d'Italia, Temi di discussione 792



Questions?

mcaruso4@os.uniroma3.it · https://pel.carusomatteo.it